Wire Strander
Fall 2024 ME Capstone Executive Summary

Design Team
Luke Bottger, Cody Clifford, Karen Hamaguchi, Kendrick Langenbach, Ted Yee

Design Advisor
Professor Scott Julien,
Mechanical and Industrial Engineering, Northeastern University

Abstract

Northeastern University's Additive Manufacturing Lab has partnered with the undergraduate capstone program and requisitioned our team of students to support their research of atypical alloys in wire arc additive manufacturing (WAAM). In order to aid them, we have created a wire stranding machine that twists together seven wires of the user's choice into a WAAM feedstock that melts into a homogenous alloy during deposition. To educate our design decisions, we began with research into WAAM technology, including physical constraints and existing applications, as well as existing industrial stranders. We focused on size, safety, and adjustability and created an enclosed benchtop device that twists standard-size spools with adjustable pitch and tension. Throughout the process, we iterated major design points such as our tensioning system, motor specifications, and types of input materials. Ultimately, we have created a device capable of producing uniformly pitched, concentric wires that can be fed through a WAAM machine and used as feedstock. The mixed composition wires that our strander produces can benefit WAAM research by quickly producing highly custom feedstock that allows use of WAAM with new materials, particularly high-entropy alloys.

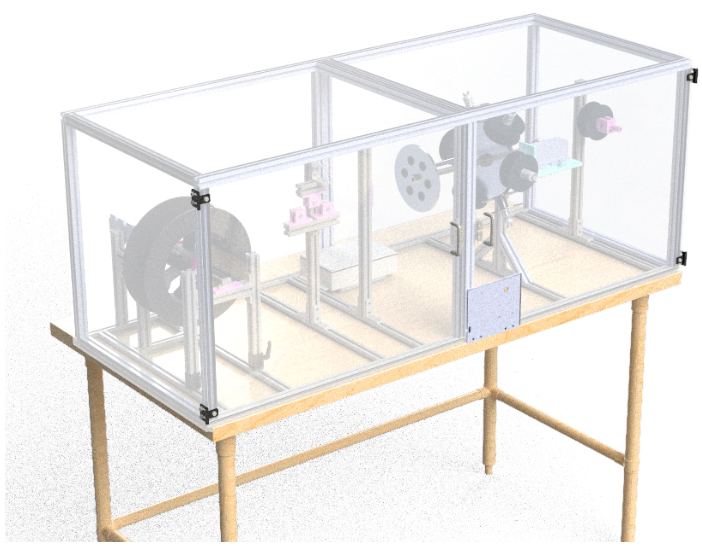

Figure 1: CAD model of the final design of our wire strander

For more information, please contact s.julien@northeastern.edu.

Need

With this project, we aim to enable wire arc additive manufacturing (WAAM) to use a wider variety of materials, especially high entropy alloys (HEA's). Professor Ahmad Nourian Avval, who researches additive manufacturing at the Northeastern University Mechanical Engineering Department, proposed this project to expand the capabilities of a standard WAAM cell from printing with only materials sold in a solid wire feedstock to printing materials that cannot be drawn into a solid wire. This will be done by stranding together a feedstock 'cable' out of component wires. When this feedstock passes the welding torch in the WAAM cell, the component wires melt together and form an alloy during deposition. If successful, this method is ideal for creating materials like HEA's. HEA's have better mechanical properties like fracture toughness, hardness, and wear resistance than many other materials but are difficult to manufacture [1], [2]. Successful use of our wire strander could allow WAAM cells, like the one used in Northeastern's research, to create parts with new, diverse materials 'in-house.'

Background and Significant Prior Work

This section explores the foundations of our product, which guided our design decisions and goals for the wire strander. We explain the background of WAAM technology, existing wire stranding solutions, and relevant material properties, like the yield strengths of the wires. This research gave us insight into how to create a uniform stranded wire that is suitable for WAAM applications and the appropriate forces to put on the wires during stranding. These concepts in turn influenced our decision to prioritize adjustable motor speeds and tensioning to accommodate for wires of different materials.

WAAM Technology

WAAM is a type of additive manufacturing defined by ISO/ASTM Standard 52900:2021 as a type of additive manufacturing where material is fused in layers by directing energy to a specific location [3], [4]. Simply put, it involves depositing metal layers on top of each other, creating a 3D part. As wire feedstock is fed through the machine, a welding torch on a robot arm melts the wire onto a substrate. To fit the constraints of the WAAM cell our wire will interact with, the outer diameter of the stranded wire must fall between 0.732mm and 3.302mm. Additionally, the wire must be fed through 10ft of flexible tubing without unraveling or damaging the tubing. To ensure that wires with a wide range of diameters and strengths can be used, we have prioritized adjustable tensioning and motor speeds as the wire is stranded.

In a 2020 research paper [5], published in the Journal of Materials Science and Technology, the authors suggest that stranded wire can be used in WAAM to create high-entropy alloys. These alloys showed high strength and ductility, and the mechanical properties of the material can be influenced by adjusting the speed of the torch. This highlights the importance of our client's research, as it has major implications for many industries, including aerospace, defense, electronics, and medicine.

Existing Solutions

Today, the largest application of wire stranding involves making cables with large-scale machines, as shown in Figure 2. However, no convenient solutions exist to address the small-scale, varying nature of material science research. Our client's research requires a relatively small, affordable, and easy to use wire strander that is adjustable so it can be used with different wire combinations.

Large industrial-size wire stranding machines like the one shown in Figure 2 are prominent today. Many of these machines create wires that are too large to be fed into the welder and take up entire rooms, making them impractical for Northeastern's Additive Manufacturing Lab to keep in-house, however, these large stranders have informed many of our design decisions.

Figure 2: Industrial wire stranding machine [6]

We researched several documentaries involving the process of creating wire cable, which we used to determine the best industry practices. We adopted some concepts from these industrial machines, such as grommets lining the holes through which the wires pass, covering sharp edges and avoiding unnecessary abrasion on our wire [7], [8]. In addition, these videos showed sheet metal guards or wire cages around the machines' moving components. We have included a safety shield around our strander to ensure safety and prevent injury or interference while it is in use. Another feature that many of these stranding machines include is a set of 'bobbins' holding the input spools. The bobbins hold the spools upright as they are wound around the core wire. This avoids unnecessary stress while twisting the wires. We considered implementing a similar mechanism to our design, but upon experimentation, there was no significant amount of stress being placed on the wires without bobbins. Therefore, we opted to prioritize a more compact, user-friendly design of our strander. Examples of each of these design elements can be seen in Figure 3 below.

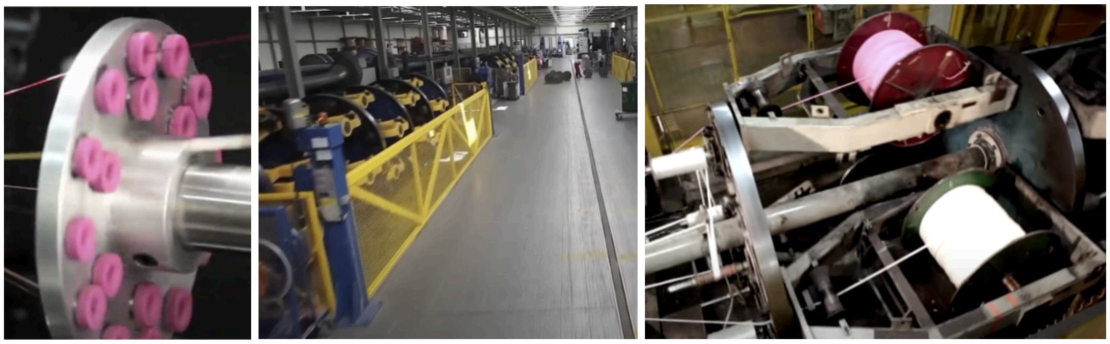

Figure 3: Stranding machine elements. Left to right: grommets [9], wire cage [10], bobbins [8]

As we began testing our wire, we found that some wires have more springback than others and are less likely to hold their shape after being stranded together. To address this, we turned to a mechanism called an over twister, which we found during patent research [11]. This idea, similar to the straightening rollers shown in Figure 4 below, helps to reduce residual forces in the wires after being stranded. They make the wire less likely to unwind, as the mechanism yields each wire of the strand together, in the same direction. Implementing this mechanism into our design helps to create consistent, tightly wound wire.

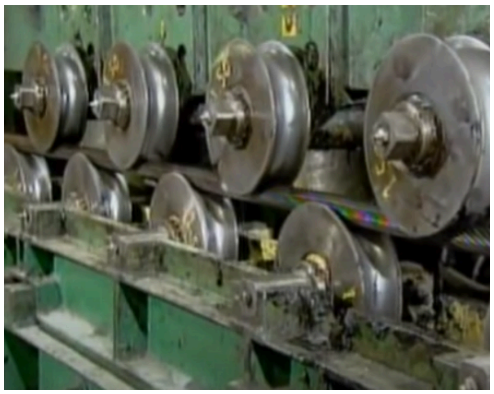

Figure 4: Straightening rollers [9]

Another source of custom stranded wire is specialty vendors, including California Wire Co. [7] and Jersey Strand and Cable [12]. These companies offer cables of 3, 7, or 19 wire strands. However, they lack the combination of material selection and wire diameter our client requires. In addition, with their own wire strander, our client can more quickly create and iterate on the wire they create without having to deal with lead times and vendor pricing. Avoiding these variables will speed up the overall research process immensely. The downfalls of buying from vendors validate the need for the client to be able to create their own feedstock.

Background Conclusion

As WAAM becomes more prominent as a 3D printing technology, Northeastern's additive manufacturing lab is looking to use it to research the fabrication and properties of high-entropy alloys. Recent research has shown that this method may be an effective way to produce these alloys, although it is currently done using highly controlled processes. As we moved through the design process, we took several factors into account including compatibility with the WAAM cell, the safety of the machine itself, and the composition of the wires. Current solutions that provide stranded wire are too large and expensive or are unavailable for purchase. With our wire strander, the additive manufacturing lab will have a desktop-sized solution that is capable of rapidly producing wire to advance their research in the metallurgy field.

Design Solution

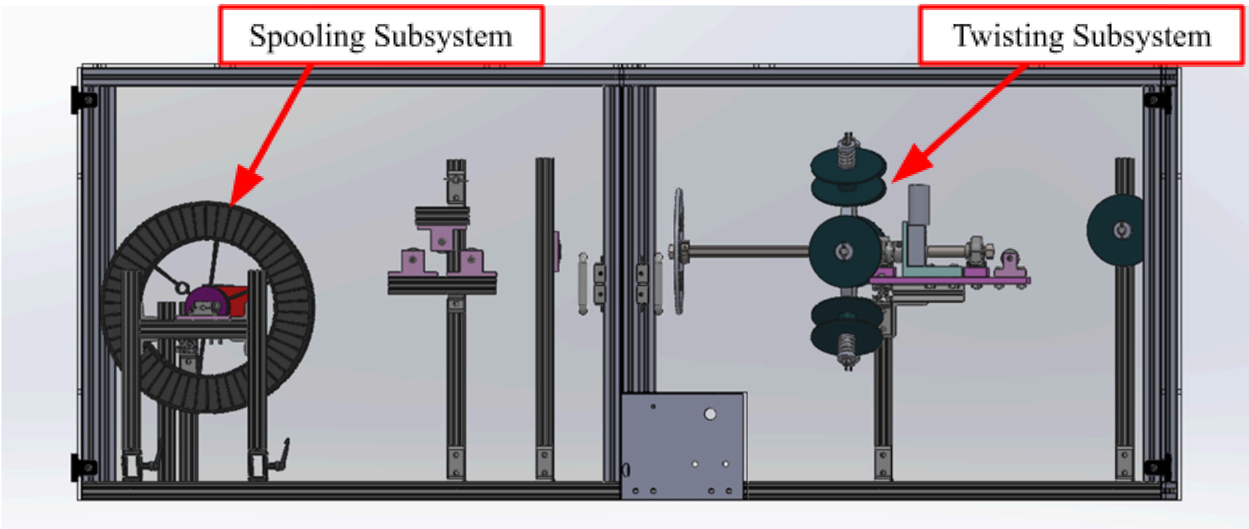

Figure 5: CAD of the final design of the stranding machine

Our final design solution, shown in Figure 5, consists of two main subsystems: the twisting subsystem and spooling subsystem. The entire machine is supported by adjustable 1” 8020 framing (T-slotted aluminum extrusion bars) and enclosed with an acrylic plexiglass safety cage protecting the user from pinch points.

The twisting subsystem, shown on the right side of Figure 5, twists the wires together and is centered around a hollow shaft. The shaft is hollow to allow a center wire to pass through, which is used as the core wire in the 7-wire concentric stranding pattern. There are two disks that are mounted on this shaft to rotate at the same rate: one disk has 6 spool mounts with individually adjustable spring-loaded tensioners, and the other has 7 holes for passing through each of the wires before joining.

As the shaft and disks are driven to rotate by an ‘input motor,’ the spooling subsystem is rotated by an ‘output motor’ at a fixed rate. The stranded wire moves at a fixed linear rate determined by the output motor, passing through a joining grommet and the overtwister mechanism onto the output spool. The output spool also makes use of a cam to oscillate along its rotational axis so that the wire gets spooled neatly in layers, similar to a fishing reel spool.

Our client needs to be able to adjust the pitch of the wire strands created to adjust the compositions of the alloys. In our final design, the pitch length can be changed via the relationship between the speeds of the input (twisting) and output (spooling) motors. The user can adjust these speeds with two simple dials on a UI panel at the front of the machine. The panel also houses buttons for ON/OFF and emergency stop. The user can load six input spools onto the spool wheel and a seventh onto the center spool support on the rightmost 8020 post. The user can also load an empty spool onto the spooling subsystem shaft and take it off after the machine has been run and the spool is full. We also included reed switches, preventing the machine from operating if either door is open.

Hollow shaft

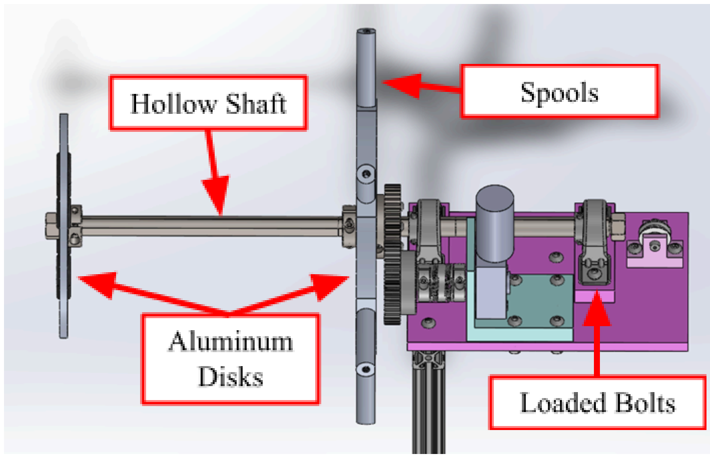

Figure 6: All parts associated with the loads and rotation of the hollow shaft

Figure 6 above shows the hardware which in combination with the input apply downward loads on the shaft like a cantilever. In our first prototype, we used a steel shaft because we did not have finalized designs for the tensioning mechanism or twisting system components. However, after finalizing our design, we performed hand calculations and FEA to determine that we could use aluminum for the hollow shaft with a factor of safety against yielding of over 100.

For this system, the lowest factor of safety against yielding is for the bolts on the first bearing that holds the hollow shaft. Using the methods from the 11th Edition of Shigley's Mechanical Engineering Design textbook [13], we found that the bolt had a yielding safety factor of about 1.5 with a high preload stress and a steel hollow shaft. As the preload stress is decreased, the safety factor increases, and when the aluminum shaft is used, the load on the bolts also decreases. Because of these changes that we were able to make, we determined that the bolts would not fail under normal operating conditions.

This shaft is rotated using a 200RPM DC motor, which rotates at 80RPM under load. We chose this motor because it has both the low- and high-end speeds necessary to create wire of 1" pitch length at a rate of 1-3 in/s. Calculations of the torque to simply accelerate the mass from rest gave confidence in the motor we chose, but quantifying other forces like friction and forces required to bend the wires was difficult so we chose to rely on testing to validate our motor selection. After testing, wires with high springback forces such as stainless steel proved to require more torque than our current input motor can provide. Simply upgrading to higher torque motors would solve this issue.

Individual Spring-loaded Tensioning System

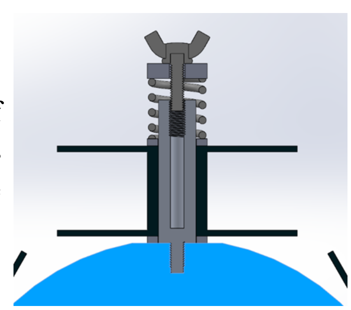

Figure 7: Cross-section of tensioning mechanism

One requirement for this machine is the use of multiple different types of metal wire, like aluminum, copper, steel, and nickel. Each input spool has its own spring-loaded tensioning mechanism, as shown in Figure 7, to ensure that each wire will yield to its final stranded shape.

The mechanisms' constructions are centered on aluminum shafts threaded into the input wheel (the main rotating disk) which each supports a spool sandwiched between two plastic disks. These disks are essential for providing tension to individual wires. This top disk supports a spring compressed by an aluminum cap and a wing nut which threads into the mounting axle. Twisting the wing nut increases compression in the spring which causes plastic disks to impart more clutch friction on the spool, keeping the wire taut during operation.

Both the threads between the wheel and shaft and the shaft and wingnut will be subject to variable magnitudes of weight, centripetal force, spring compression, and shear stress as the input wheel rotates. As such, it was necessary to calculate factors of safety for excessive loading, joint separation, and fatigue under normal operating circumstances [13]. The lowest factors of safety we found for each thread were 9.6 for the fully compressed wing nut and 8.2 for the mounting axle. Given these safety factors and the fact that the device cannot operate at very high speeds, we are confident that these particular elements will not fail and that the spools will not come loose.

<ins>Post-twist processing of the wire strand</ins>

In our research, we found that many industrial machines use pulley systems to yield the stranded wires together and remove any impressions left from the spools they were on while maintaining the pitch length. We adopted an over twister mechanism from industrial machines, shown in Figure 8, that can be adjusted to yield the wires more or less, as required, depending on the height of the middle pulley. We can adjust the height by changing the height of the 8020 frame to which it is mounted.

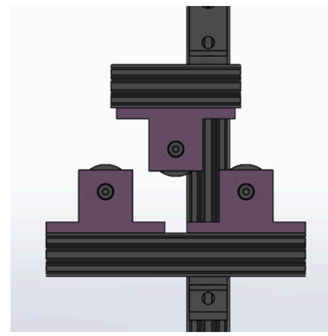

Figure 8: Overtwister system made of pulleys

<ins>Output spooling oscillation</ins>

We purchased a 90-degree gearbox, shown in Figure 9 under the transparent dust cap, which has a gear ratio of 20:1, allowing us to evenly wrap the wires along the width of the spool. As the spool rotates with the shaft, the cam, shown in purple, attaches to parts sandwiching the spool and oscillates it along the length of a ball spline. A ball spline is a specially designed shaft that allows rotation and linear motion with little friction. One important feature of this design is that the spool can be mounted and removed as the user desires. This is done by unscrewing the two screws connecting the spool caps together, shown in Figure 10. The light purple side has chamfered screw holes so that the screws can be guided toward them easily by feel.

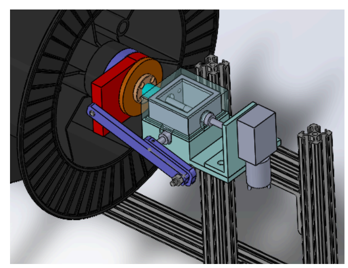

Figure 9: 90-degree gearbox

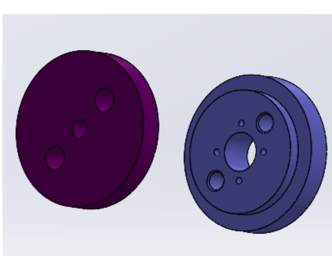

Figure 10: Caps fit to standard welding spool

The motor used to spool the wire is rated for 5RPM and 35 kg-cm of torque. Running at about 2.5RPM for our desired pitch length, the motor has a torque factor of safety of 6 before the inclusion of friction and other errors. When we first selected a motor for this, we chose a motor with too little torque that was not able to spin the loaded spool, oscillate it for even spooling, and pull the wire tension. Our only non-compromising solution was to purchase a new motor.

Safety and Electronics

We chose to make our safety cage out of clear acrylic panels to keep the user safe and allow them to observe the operation. This protects the user from pinch points created by rotating and converging wires and protects components from dust. In the case that the wires break, an acrylic shield will also protect the user better than a wire cage.

We designed our electronics around safety considerations, prioritizing simple designs and plug-and-play components. Power from a 120V wall outlet passes through the ON/OFF buttons, into an AC/DC 12V power converter, and through a relay. The relay can be interrupted by opening either door or by pushing the emergency stop button. Power then passes through two separate motor controllers (with dials on the front UI panel) and into each motor. We ensured no access to high voltage contact and multiple safety kill switches, while strain-relieving and concealing all wires.

Design Solution Summary

We were able to create a functional and flexible design for our wire strander. It can successfully take any wire on a common spool size, create a stranded 'cable' with variable pitch and tension, and spool it onto a spool for use with WAAM. The user is protected from electrical, impact, and pinch point injuries during operation. We ensured that the entire machine is modular through removable parts and adjustable 8020 parts, allowing repairs and adjustments in as many ways as possible.

Design Process

We primarily designed around the type of strand we wanted to create. To satisfy client requirements, we chose a seven-wire concentric strand (shown in Figure 11), consisting of one central wire with six outer wires wrapped around the central wire. We also considered a nineteen-wire concentric strand, non-concentric strands, and woven wire strands, but we chose the seven-wire concentric strand for the overall size of the wire, simplicity, and its similarity to solid wire.

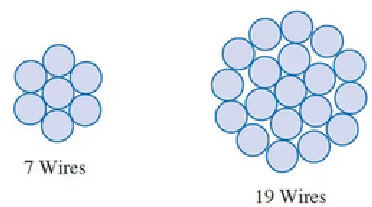

Figure 11: Concentric Stranding Pattern

By choosing a seven-wire concentric strand, we limited our geometry in several ways. On the input side of the device, we needed to accommodate one "core" wire that runs through the center of the device and does not twist. Next, we needed to find a way to wrap the other six input wires around the central wire and create a system to apply equal tension to all the wires for even wrapping. In parallel, we also needed a system to feed the stranded wire onto an output spool for use in the WAAM machine. We weighed several methods to accomplish the various tasks of each system in depth, but the general purpose of each subsystem was constrained by the stranding method we chose.

From the start, we aimed to design quickly and iterate often. We wanted to make mistakes fast and learn from them. We quickly had parts ready to be manufactured and continued to submit parts for manufacturing throughout the semester. Building functioning prototypes with these parts proved to be a challenge with different members working on different subsystems and part changes happening quickly.

However, by having these parts in hand, we were able to easily identify modifications that needed to be made.

We tested as frequently as we had new parts throughout the semester. As parts became more polished, our wire-stranding tests similarly became more meaningful. Initially, we twisted wire by hand, and while we had no device to test, it helped us understand the properties of our wire and how it behaved. We later conducted tests with assembled components but with no motors, allowing us to have a physical feel of how the wire twisted. We tested early motor-powered prototypes with seven aluminum wires, allowing us to see where weaker metals would break, but not testing the limits of the machine's power. In our final design, we are able to twist a high-quality wire with our machine, test different materials, and control parameters such as tension and pitch length to strand an ideal wire. We felt most confident in the success of our design when building our final prototype. Key factors, such as a more complex tensioning system, an updated input wire system, and machined parts, made this version much more operable than previous builds.

Originally, we divided tasks by subsystem, but through weekly advisor meetings and regular build times, every member shared their progress and handed off tasks when additional help was needed. Whether it was CAD, building, or offering feedback, our design represents the entire group's ideas. This fluid sharing of tasks led to a stronger design than if tasks had been strictly divided.

Results

Having completed prototypes that can create wire without human interference, we have been able to conduct testing with three metals of different strengths (aluminum, stainless steel, and mild steel), laying some concerns to rest, while necessitating other changes. We were pleased to note that the created wires were incredibly uniform in pitch and diameter, with no noticeable variation in the length between subsequent twists and a profile that appeared to be as circular as possible given the design limitations.

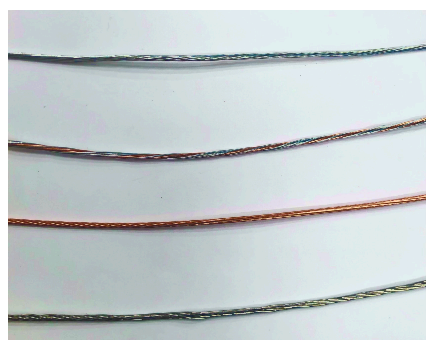

Figure 12: Stranded wire samples containing various combinations of 3034Al, er70S, and 316L SS

Notably, the completed aluminum prototype wire holds together well as it is spun onto the output spool. Due to the combination of the input spool tensioning mechanisms, the over-twister, and a short pitch length, the aluminum wires yield enough to retain their relative position to one another without significantly deforming the overall strand. This has also held true as well for mild steel wires, however, when stranding only mild steel, the springback forces of the combined wire cause our current input motor to stall. A higher quality mild steel strand could be created when powering the input spool by hand, demonstrating that stronger motors would resolve this issue. One of our primary concerns throughout this project has been the wire coming undone while feeding through the WAAM cell tubing, which would require maintenance and disassembly. While this is still a concern, the success of the wires staying together so far have shown that a properly stranded wire will be able to travel to the plasma welder nozzle intact under typical handling and machine loading.

One way we've tested the feedability of our wire is with a wire feeder, intended to be used with a welder, which we have purchased for testing. Testing on this wire feeder has been a good indication of how our wire will interact with the similar wire feeder in the WAAM cell, however, the wire feeder that we have tested with does not have rollers for the large diameter of our stranded wire, and we hope that the issues the smaller rollers have caused could be fixed with larger rollers. Initial testing with our 7-wire aluminum strand resulted in small amounts of flattening and deformation of the wire; however, the wire still held together after being stranded. While mixing aluminum and mild steel wires also could not feed, wire made of all seven mild steel wires was successful with the small rollers. The mixed wire is the most representative combination of the types of wire that this device will often be used to make.

Some of the final components to be integrated into our device were the door switches and emergency stop. These work as intended; the machine will turn off when either door is open or the emergency stop button is depressed. Our motors and their controllers are also working well, capable of changing the production rate and pitch of the wire. This allows for greater flexibility in the continuous operation of the stranding device.

Summary and Impact

WAAM is a burgeoning technology that promises to innovate the manufacturing of precision metallic components. While Northeastern University is in the early stages of experimenting with this technology, their research will help the world better understand its possibilities and limitations. We hope that the successful construction of the wire strander will expand the university's material capabilities and accelerate improvement of WAAM as a whole.

Understanding the mechanics behind wire twisting was initially difficult due to a lack of literature on the process. Generally, companies have not shared information on their technologies which left us to learn from only patents and a few research papers. With that information and by working quickly towards initial prototypes we allowed ourselves time to observe the mechanical properties of the wires and make informed design decisions through them. This approach paved a path for us from early design and ideation to the functioning prototype we have today. Our device has proven it is capable of combining up to seven wires together in a secure, uniform package with minimal operator input. Both 7 aluminum and combinations of aluminum and mild steel wires have been successfully twisted with our wire strander. The current largest drawback of our device is an underpowered input motor, incapable of twisting 7 high springback wires, such as stainless steel. However, our success in creating a high-quality twist by hand gives us confidence that the same quality twist of steel can be created by the motor driven device.

After Capstone Day, we plan to have the wire strander transported to the Burlington campus to assist with WAAM research. We hope this device can assist their research and allows the creation of parts with additional alloys. We are pleased with the outcome of our project but see room for improvement in our device including adding higher torque motors and making it easier to start a new strand of wire.

References

[1] M. Tsai and J. Yeh, "High-Entropy Alloys: A Critical Review," 2014. Available: https://www.tandfonline.com/doi/full/10.1080/21663831.2014.912690.
[2] J. Yeh *et al*, "Nanostructured high‑entropy alloys with multiple principal elements: novel alloy design concepts and outcomes," *Advanced Engineering Materials,* vol. 6, *(5),* pp. 299–303, 2004.
[3] *Shen Qingkai, a postgraduate student published a cover paper in the international journal of metal materials*. Available: https://www.wzu.edu.cn/info/1465/33536.htm.
[4] Q. Shen, X. Kong and X. Chen, "Fabrication of bulk Al-Co-Cr-Fe-Ni high-entropy alloy using combined cable wire arc additive manufacturing (CCW-AAM): Microstructure and mechanical properties," *Journal of Materials Science & Technology,* vol. 74, pp. 136–142, 2021. Available: https://www.sciencedirect.com/science/article/pii/S1005030220309063. DOI: 10.1016/j.jmst.2020.10.037.
[5] *A Short History of Wire Rope*. Available: https://www.ropetechnology.com/downloads/brochures/bro_a-short-history-of-wire-rope.pdf.
[6] *FLYMCA Stranding Machine*. Available: https://www.flymca.com/en/rigid-stranding-machine/.
[7] *Stranded & Twisted Wire*. Available: https://calfinewire.com/products-services/stranded-wire/.
[8] *General Cable - How Wire & Cable is Made Video*. Available: https://www.youtube.com/watch?v=hLkyUQ0B8Dw
[9] Discovery Channel, "How It's Made - Industrial Wire Ropes," September 21, 2011.
[10] *SECRETS Of The Modern Wire Rope Production Process - Giant Steel Wire Rope Manufacturing Process*. Available: https://www.youtube.com/watch?v=VBs-MKI-Xx8.
[11] "Apparatus and method for making a helically wound conductor," Goodyear Tire and Rubber Co and James Christopher Kish, , January 14, 2010.
[12] *Stranded Wire*. Available: https://jerseystrandandcable.com/custom-stranded-proprietary-wire/.
[13] R. G. &. Budynas and J. K. Nisbett, *Shigley's Mechanical Engineering Design.* (11th ed.) McGraw-Hill Education, 2020.

Appendix: Engineering Analysis

For our Capstone project, we are developing a wire strander. The aim of the device is to twist together 7 metal wires together from individual 'input' spools to produce one uniform stranded wire that is wound onto an 'output' spool. This wire is then intended to be fed into a wire-arc manufacturing (WAAM) machine, a device that uses a welder to 3D print metallic parts. Some of the most important components of this design are two bearings that support a hollow rotating central shaft holding the input spools and guiding parts. We resolved to find factors of safety for the bolts supporting both bearings, namely the factors against yielding and joint separation. The calculations for the bolts on the bearing furthest from the input spool are shown here, as they are the most likely point of failure in our entire assembly. We wrote a MATLAB script to calculate the yielding and joint separation safety factors for the bolts using Shigley's methods. With an SAE grade of 5 and a preload stress set to 75% of the proof strength, these bolts have a yielding safety factor of 1.33 and a separation safety factor of 400. We determined that these safety factors are acceptable, but if we want to increase the yielding safety factor, we can decrease the preload stress.

To calculate these safety factors, we first calculated the grip length, threaded length, unthreaded grip length, threaded grip length, unthreaded area, and fastener stiffness of the bolt using the suggested procedure from Table 8-7. We determined the grip length to be the thickness of all materials squeezed between the face of the bolt and the face of the nut. We used Equation 1 below:

$$l = t_1 + t_2 + t_3 + t_w \qquad \text{(Eq. 1)}$$

where *l* is the grip length, *t_1* is the thickness of the mounting plate, *t_2* is the thickness of the height adjustment plate for the bearing, *t*_3 is the bearing mounting plate, and *t_w* is the thickness of the washer, all in inches.

The bolt that we used for fastening the bearings down are ¼"-20 screws of 1.375 in length. Because the overall length of the screw is less than 6 inches, the equation we used for the threaded length was:

$$L_T = 2d + \frac{1}{4} \qquad \text{(Eq. 2)}$$

where *L_T* is the threaded length and *d* is the diameter of the screw (1/4"), both in inches. The unthreaded grip length was calculated using Equation 3,

$$l_d = L - L_T \qquad \text{(Eq. 3)}$$

where *l_d* is the unthreaded grip length in inches, and *L* is the screw length in inches. The threaded grip length was calculated from Equation 4,

$$l_t = l - l_d \qquad \text{(Eq. 4)}$$

Meanwhile, the unthreaded area was calculating using,

$$A_d = \frac{\pi d^2}{4} \qquad \text{(Eq. 5)}$$

where *A_d* is the unthreaded portion area in square inches, and *d* is the diameter in inches. With all calculable variables accounted for, we could then find the fastener stiffness using the equation,

$$k_b = \frac{A_d A_t E_{steel}}{A_d l_t + A_t l_d} \qquad \text{(Eq. 6)}$$

where *k_b* is the fastener stiffness in lbf/in, *A_t* is the threaded portion area found from Table 8-2, and *E_steel* is the Young's Modulus of the steel bolt in psi, which is about 30 million psi.

We performed the next part of the screw calculations using frusta calculations. Frusta are trapezoidal representations of the spread of compression force through various materials. The mounting plate and the height adjustment bearing mount are both aluminum and the bearing mounting plate and washer are made of steel. Figure 2 below shows the thicknesses and outer diameters used in the stiffness calculations.

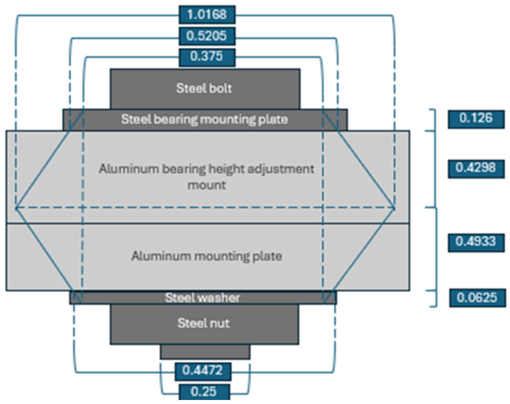

Figure 2: 30° frusta diagram used to calculate member spring rate

We calculated the distance *t_frusta* that the frusta extend by dividing the overall thickness held together by the screw and nut by 2:

$$t_{frusta} = \frac{t_1+t_2+t_3+t_w}{2} = \frac{l}{2} \tag{Eq. 7}$$

For this bolt, we determined that there were four frusta: the steel bearing mounting plate frustum, the upper aluminum frustum, the lower aluminum frustum (which contains both the bottom section of the aluminum bearing height adjustment mount and the aluminum mounting plate), and the steel washer frustum. We calculated the outer diameters *D* of the frusta of the steel bearing mounting plate and washer using:

$$D_{steel} = 1.5 * d \tag{Eq. 8}$$

The next few values of *D* required more in-depth calculations. We calculated the outer diameter *D_upper* of the upper aluminum frustum using Equation 9:

$$D_{upper} = 2t_3 * \tan(30) + D_{steel} \tag{Eq. 9}$$

Then we calculated the outer diameter of the lower aluminum frustum with Equation 10:

$$D_{lower} = 2t_w * \tan(30) + D_{steel} \tag{Eq. 10}$$

Finally, we could calculate the spring rates of each part using the equation:

$$k = \frac{0.5774\pi E d}{ln\left[\frac{(1.155t+D-d)(D+d)}{(1.155t+D-d)(D+d}\right]} \tag{Eq. 11}$$

where *E* is the Young's Modulus of the frustum, *d* is the screw diameter, *t* is the thickness of the frustum, and *D* is the outer diameter of the frustum.

The thicknesses of the steel bearing mounting plate and washer are the thicknesses of each of those parts.

However, we had to calculate the individual thicknesses of the aluminum frusta, given their components had differing thicknesses. We calculated the upper aluminum frustum thickness using Equation 12,

$$t_{upper} = t_{frusta} - t_3 \qquad \text{(Eq. 12)}$$

We then proceeded to calculate the lower aluminum frustum thickness using Equation 13,

$$t_{lower} = t_{frusta} - t_w \qquad \text{(Eq. 13)}$$

This was followed by the calculation for the spring rate of the steel bearing mounting plate using:

$$k_{steelPlate} = \frac{0.5774\pi E_{steel} d}{ln\left[\frac{(1.155t_3 + D_{steel} - d)(D_{steel} + d)}{(1.155t_3 + D_{steel} - d)(D_{steel} + d}\right]} \qquad \text{(Eq. 14)}$$

We then calculated the spring rate of the steel bearing mounting plate using:

$$k_{upperAl} = \frac{0.5774\pi E_{Al} d}{ln\left[\frac{(1.155t_{upper} + D_{upper} - d)(D_{upper} + d)}{(1.155t_{upper} + D_{upper} - d)(D_{upper} + d}\right]} \qquad \text{(Eq. 15)}$$

where *E_Al* is the Young's Modulus of aluminum. We performed the following frusta calculation of the steel bearing mounting plate using:

$$k_{lowerAl} = \frac{0.5774\pi E_{Al} d}{ln\left[\frac{(1.155t_{lower} + D_{lower} - d)(D_{lower} + d)}{(1.155t_{lower} + D_{lower} - d)(D_{lower} + d}\right]} \qquad \text{(Eq. 16)}$$

The final calculation for the spring rate of the steel bearing mounting plate was found using the following equation:

$$k_{steelWasher} = \frac{0.5774\pi E_{steel} d}{ln\left[\frac{(1.155t_w + D_{steel} - d)(D_{steel} + d)}{(1.155t_w + D_{steel} - d)(D_{steel} + d}\right]} \qquad \text{(Eq. 17)}$$

The overall spring constant, *k_m*, could finally be calculated using the below summation formula:

$$k_m = \frac{1}{\frac{1}{k_{steelPlate}} + \frac{1}{k_{upperAl}} + \frac{1}{k_{lowerAl}} + \frac{1}{k_{steelWasher}}} \qquad \text{(Eq. 18)}$$

It was then necessary to find the stiffness constant of the joint *C*, calculated using Equation 19:

$$C = \frac{k_b}{k_b + k_m} \qquad \text{(Eq. 19)}$$

We then used the Bowman recommendation to calculate the preload force *F_i*:

$$F_i = 0.75 * \sigma_{proof} * A_t \qquad \text{(Eq. 20)}$$

where *σ_proof* is the proof strength, found in Table 8-9. For these calculations, we assumed an SAE grade of 5 for each bolt. We then calculated the resultant bolt stress using Equation 21:

$$\sigma_{bolt} = \frac{CF_{load} + F_i}{A_t} \qquad \text{(Eq. 21)}$$

where *F_load* is the tensile load experienced by each bolt in lbf.

We could finally find the factors of safety for the bearing bolts. The factor of safety against yielding is presented as Equation 22:

$$N_{yield} = \frac{\sigma_{proof}}{\sigma_{bolt}} \qquad \text{(Eq. 22)}$$

Meanwhile, the safety factor against separation is given by Equation 23:

$$N_{separation} = \frac{F_i}{F_{load}(1-C)} \quad \text{(Eq. 23)}$$

Following careful analysis of the system and the forces acting on it, the factor of safety against yielding was found to be 1.33, while the factor of safety against joint separation was 400. While the yielding factor was found to be somewhat low, the limited range of speed our device is capable of running at gives us faith that the second bearing should not fail under normal operation.